

*****Draft*****

pico & pico2
spi-master-slave
08/07/26

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Two Raspberry Pi 5 8Gb systems were used in developing the code.

Testing the original spi_master_slave code.

```
devel@pi5-80:~ $ mkdir spi
```

```
devel@pi5-80:~ $ cd spi/
```

```
devel@pi5-80:~ $ git clone -b test-dev https://github.com/develone/pico-examples.git
```

```
devel@pi5-80:~/spi $ cd pico-examples/
```

```
devel@pi5-80:~/spi/pico-examples $ git checkout 314d
```

```
devel@pi5-80:~/spi/pico-examples $ git log
commit 314d551230e637ba111ab14c7e4fcb73ff9ac114 (HEAD)
Merge: 026f37b 4c3a3dc
Author: develone <develone@sbcglobal.net>
Date: Thu Aug 7 06:24:36 2025 -0600
```

Merge branch 'raspberrypi:master' into test-dev

This was the Raspberry Pi 5 8Gb pi5-80 build which is the original code for the spi_master_slave with spi/spi_master_slave/spi_slave/CMakeLists.txt modified.

```
devel@pi5-80:~/spi/pico-examples $ git diff spi/spi_master_slave/spi_slave/CMakeLists.txt
```

```
diff --git a/spi/spi_master_slave/spi_slave/CMakeLists.txt
```

```
b/spi/spi_master_slave/spi_slave/CMakeLists.txt
```

```
index 8e05645..6f83582 100644
```

```
--- a/spi/spi_master_slave/spi_slave/CMakeLists.txt
```

```
+++ b/spi/spi_master_slave/spi_slave/CMakeLists.txt
```

```
@@ -1,12 +1,20 @@
```

```
-add_executable(spi_slave
```

```
-    spi_slave.c
```

```
+if (TARGET tinyusb_device)
```

```
+    add_executable(spi_slave
```

```
+        spi_slave.c
```

```
)
```

```
-# Pull in basic dependencies
```

```
-target_link_libraries(spi_slave pico_stdlib hardware_spi)
```

```
+    # Pull in basic dependencies
```

```
+    target_link_libraries(spi_slave pico_stdlib hardware_spi)
```

```
-# create map/bin/hex file etc.
```

```
-pico_add_extra_outputs(spi_slave)
```

```
+    # create map/bin/hex file etc.
```

```

-# add url via pico_set_program_url
-example_auto_set_url(spi_slave)
+
+    pico_add_extra_outputs(spi_slave)
+    pico_enable_stdio_usb(spi_slave 1)
+    pico_enable_stdio_uart(spi_slave 0)
+
+    # add url via pico_set_program_url
+    example_auto_set_url(spi_slave)
+elseif(PICO_ON_DEVICE)
+    message("Skipping hello_usb because TinyUSB submodule is not initialized in the SDK")
+endif()

```

This is what is seen on the minicom myusb0 when the pico2 spi_master is power-up.

SPI master example

SPI master says: The following buffer will be written to MOSI endlessly:

```

00 01 02 03 04 05 06 07 08 09 0a 0b 0c 0d 0e 0f
10 11 12 13 14 15 16 17 18 19 1a 1b 1c 1d 1e 1f
20 21 22 23 24 25 26 27 28 29 2a 2b 2c 2d 2e 2f
30 31 32 33 34 35 36 37 38 39 3a 3b 3c 3d 3e 3f
40 41 42 43 44 45 46 47 48 49 4a 4b 4c 4d 4e 4f
50 51 52 53 54 55 56 57 58 59 5a 5b 5c 5d 5e 5f
60 61 62 63 64 65 66 67 68 69 6a 6b 6c 6d 6e 6f
70 71 72 73 74 75 76 77 78 79 7a 7b 7c 7d 7e 7f
80 81 82 83 84 85 86 87 88 89 8a 8b 8c 8d 8e 8f
90 91 92 93 94 95 96 97 98 99 9a 9b 9c 9d 9e 9f
a0 a1 a2 a3 a4 a5 a6 a7 a8 a9 aa ab ac ad ae af
b0 b1 b2 b3 b4 b5 b6 b7 b8 b9 ba bb bc bd be bf
c0 c1 c2 c3 c4 c5 c6 c7 c8 c9 ca cb cc cd ce cf
d0 d1 d2 d3 d4 d5 d6 d7 d8 d9 da db dc dd de df
e0 e1 e2 e3 e4 e5 e6 e7 e8 e9 ea eb ec ed ee ef
f0 f1 f2 f3 f4 f5 f6 f7 f8 f9 fa fb fc fd fe ff
SPI master says: read page 0 from the MISO line:
fb fa f9 f8 f7 f6 f5 f4 f3 f2 f1 f0 ef ee ed ec
eb ea e9 e8 e7 e6 e5 e4 e3 e2 e1 e0 df de dd dc
db da d9 d8 d7 d6 d5 d4 d3 d2 d1 d0 cf ce cd cc
cb ca c9 c8 c7 c6 c5 c4 c3 c2 c1 c0 bf be bd bc
bb ba b9 b8 b7 b6 b5 b4 b3 b2 b1 b0 af ae ad ac
ab aa a9 a8 a7 a6 a5 a4 a3 a2 a1 a0 9f 9e 9d 9c
9b 9a 99 98 97 96 95 94 93 92 91 90 8f 8e 8d 8c
8b 8a 89 88 87 86 85 84 83 82 81 80 7f 7e 7d 7c
7b 7a 79 78 77 76 75 74 73 72 71 70 6f 6e 6d 6c
6b 6a 69 68 67 66 65 64 63 62 61 60 5f 5e 5d 5c
5b 5a 59 58 57 56 55 54 53 52 51 50 4f 4e 4d 4c
4b 4a 49 48 47 46 45 44 43 42 41 40 3f 3e 3d 3c
3b 3a 39 38 37 36 35 34 33 32 31 30 2f 2e 2d 2c
2b 2a 29 28 27 26 25 24 23 22 21 20 1f 1e 1d 1c
1b 1a 19 18 17 16 15 14 13 12 11 10 0f 0e 0d 0c
0b 0a 09 08 07 06 05 04 03 02 01 00 00 00 00 00

```

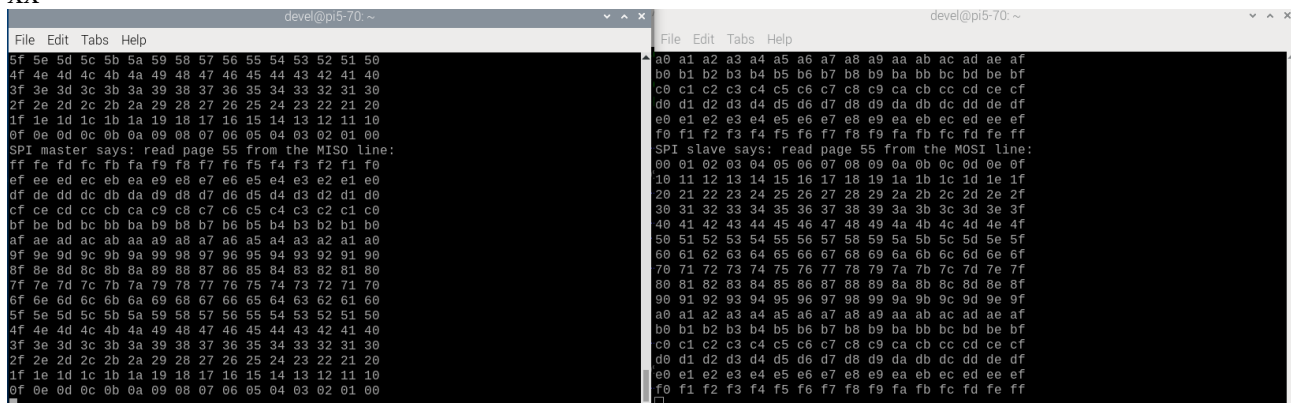
This is what is seen on the minicom myACM0 when the pico2 spi_slave is power-up.

SPI slave example

SPI slave says: When reading from MOSI, the following buffer will be written to MISO:

```
ff fe fd fc fb fa f9 f8 f7 f6 f5 f4 f3 f2 f1 f0
ef ee ed ec eb ea e9 e8 e7 e6 e5 e4 e3 e2 e1 e0
df de dd dc db da d9 d8 d7 d6 d5 d4 d3 d2 d1 d0
cf ce cd cc cb ca c9 c8 c7 c6 c5 c4 c3 c2 c1 c0
bf be bd bc bb ba b9 b8 b7 b6 b5 b4 b3 b2 b1 b0
af ae ad ac ab aa a9 a8 a7 a6 a5 a4 a3 a2 a1 a0
9f 9e 9d 9c 9b 9a 99 98 97 96 95 94 93 92 91 90
8f 8e 8d 8c 8b 8a 89 88 87 86 85 84 83 82 81 80
7f 7e 7d 7c 7b 7a 79 78 77 76 75 74 73 72 71 70
6f 6e 6d 6c 6b 6a 69 68 67 66 65 64 63 62 61 60
5f 5e 5d 5c 5b 5a 59 58 57 56 55 54 53 52 51 50
4f 4e 4d 4c 4b 4a 49 48 47 46 45 44 43 42 41 40
3f 3e 3d 3c 3b 3a 39 38 37 36 35 34 33 32 31 30
2f 2e 2d 2c 2b 2a 29 28 27 26 25 24 23 22 21 20
1f 1e 1d 1c 1b 1a 19 18 17 16 15 14 13 12 11 10
0f 0e 0d 0c 0b 0a 09 08 07 06 05 04 03 02 01 00
SPI slave says: read page 0 from the MOSI line:
00 01 02 03 04 05 06 07 08 09 0a 0b 0c 0d 0e 0f
10 11 12 13 14 15 16 17 18 19 1a 1b 1c 1d 1e 1f
20 21 22 23 24 25 26 27 28 29 2a 2b 2c 2d 2e 2f
30 31 32 33 34 35 36 37 38 39 3a 3b 3c 3d 3e 3f
40 41 42 43 44 45 46 47 48 49 4a 4b 4c 4d 4e 4f
50 51 52 53 54 55 56 57 58 59 5a 5b 5c 5d 5e 5f
60 61 62 63 64 65 66 67 68 69 6a 6b 6c 6d 6e 6f
70 71 72 73 74 75 76 77 78 79 7a 7b 7c 7d 7e 7f
80 81 82 83 84 85 86 87 88 89 8a 8b 8c 8d 8e 8f
90 91 92 93 94 95 96 97 98 99 9a 9b 9c 9d 9e 9f
a0 a1 a2 a3 a4 a5 a6 a7 a8 a9 aa ab ac ad ae af
b0 b1 b2 b3 b4 b5 b6 b7 b8 b9 ba bb bc bd be bf
c0 c1 c2 c3 c4 c5 c6 c7 c8 c9 ca cb cc cd ce cf
d0 d1 d2 d3 d4 d5 d6 d7 d8 d9 da db dc dd de df
e0 e1 e2 e3 e4 e5 e6 e7 e8 e9 ea eb ec ed ee ef
f0 f1 f2 f3 f4 f5 f6 f7 f8 f9 fa fb fc fd fe ff
```

XX



```
devel@pi5-70: ~  
File Edit Tabs Help  
5f 5e 5d 5c 5b 5a 59 58 57 56 55 54 53 52 51 50  
4f 4e 4d 4c 4b 4a 49 48 47 46 45 44 43 42 41 40  
3f 3e 3d 3c 3b 3a 39 38 37 36 35 34 33 32 31 30  
2f 2e 2d 2c 2b 2a 29 28 27 26 25 24 23 22 21 20  
1f 1e 1d 1c 1b 1a 19 18 17 16 15 14 13 12 11 10  
0f 0e 0d 0c 0b 0a 09 08 07 06 05 04 03 02 01 00  
SPI master says: read page 55 from the MISO line:  
ff fe fd fc fb fa f9 f8 f7 f6 f5 f4 f3 f2 f1 f0  
ef ee ed ec eb ea e9 e8 e7 e6 e5 e4 e3 e2 e1 e0  
df de dd dc db da d9 d8 d7 d6 d5 d4 d3 d2 d1 d0  
cf ce cd cc cb ca c9 c8 c7 c6 c5 c4 c3 c2 c1 c0  
bf be bd bc bb ba b9 b8 b7 b6 b5 b4 b3 b2 b1 b0  
af ae ad ac ab aa a9 a8 a7 a6 a5 a4 a3 a2 a1 a0  
9f 9e 9d 9c 9b 9a 99 98 97 96 95 94 93 92 91 90  
8f 8e 8d 8c 8b 8a 89 88 87 86 85 84 83 82 81 80  
7f 7e 7d 7c 7b 7a 79 78 77 76 75 74 73 72 71 70  
6f 6e 6d 6c 6b 6a 69 68 67 66 65 64 63 62 61 60  
5f 5e 5d 5c 5b 5a 59 58 57 56 55 54 53 52 51 50  
4f 4e 4d 4c 4b 4a 49 48 47 46 45 44 43 42 41 40  
3f 3e 3d 3c 3b 3a 39 38 37 36 35 34 33 32 31 30  
2f 2e 2d 2c 2b 2a 29 28 27 26 25 24 23 22 21 20  
1f 1e 1d 1c 1b 1a 19 18 17 16 15 14 13 12 11 10  
0f 0e 0d 0c 0b 0a 09 08 07 06 05 04 03 02 01 00  
devel@pi5-70: ~  
File Edit Tabs Help  
a0 a1 a2 a3 a4 a5 a6 a7 a8 a9 aa ab ac ad ae af  
b0 b1 b2 b3 b4 b5 b6 b7 b8 b9 ba bb bc bd be bf  
c0 c1 c2 c3 c4 c5 c6 c7 c8 c9 ca cb cc cd ce cf  
d0 d1 d2 d3 d4 d5 d6 d7 d8 d9 da db dc dd de df  
e0 e1 e2 e3 e4 e5 e6 e7 e8 e9 ea eb ec ed ee ef  
f0 f1 f2 f3 f4 f5 f6 f7 f8 f9 fa fb fc fd fe ff  
SPI slave says: read page 55 from the MOSI line:  
00 01 02 03 04 05 06 07 08 09 0a 0b 0c 0d 0e 0f  
10 11 12 13 14 15 16 17 18 19 1a 1b 1c 1d 1e 1f  
20 21 22 23 24 25 26 27 28 29 2a 2b 2c 2d 2e 2f  
30 31 32 33 34 35 36 37 38 39 3a 3b 3c 3d 3e 3f  
40 41 42 43 44 45 46 47 48 49 4a 4b 4c 4d 4e 4f  
50 51 52 53 54 55 56 57 58 59 5a 5b 5c 5d 5e 5f  
60 61 62 63 64 65 66 67 68 69 6a 6b 6c 6d 6e 6f  
70 71 72 73 74 75 76 77 78 79 7a 7b 7c 7d 7e 7f  
80 81 82 83 84 85 86 87 88 89 8a 8b 8c 8d 8e 8f  
90 91 92 93 94 95 96 97 98 99 9a 9b 9c 9d 9e 9f  
a0 a1 a2 a3 a4 a5 a6 a7 a8 a9 aa ab ac ad ae af  
b0 b1 b2 b3 b4 b5 b6 b7 b8 b9 ba bb bc bd be bf  
c0 c1 c2 c3 c4 c5 c6 c7 c8 c9 ca cb cc cd ce cf  
d0 d1 d2 d3 d4 d5 d6 d7 d8 d9 da db dc dd de df  
e0 e1 e2 e3 e4 e5 e6 e7 e8 e9 ea eb ec ed ee ef  
f0 f1 f2 f3 f4 f5 f6 f7 f8 f9 fa fb fc fd fe ff
```

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Currently the pico2 is the only master slave working. The pico needs rewiring and testing.

These are variables found .bashrc on pi5-90

```
#export WIFI_SSID=
#export WIFI_PASSWORD=
export WIFI_SSID="TMOBILE-CE06"
export WIFI_PASSWORD="k64ndv2czh8"
#export PATH="/home/devel/oss-cad-suite/bin:$PATH"
#export PIPELINEC_REPO="$HOME/PipelineC_pmod-ethernet-jpeg/PipelineC"
export PIPELINEC_REPO="$HOME/PipelineC"
export PICO_EXTRAS_PATH=/home/devel/pico-extras
#export PICO_SDK_PATH=/home/devel/pico/pico-sdk
export PICO_SDK_PATH=/home/devel/pico2/pico-sdk
#export PICO_2350=1
#export FREERTOS_KERNEL_PATH=/home/devel/rp2350-freeRTOS/test-read-crc16/FreeRTOS-
Kernel
export FREERTOS_KERNEL_PATH=/home/devel/pico2/freertos-lib/FreeRTOS-Kernel
export PICO_FREERTOS=/home/devel/pico2/freertos-lib/FreeRTOS-Kernel
export PICTOOL_FETCH_FROM_GITHUB=/home/devel/pico2/BUILT_PICOTOOL
#export PATH="$HOME/.pyenv/bin:$HOME/local/openocd/bin:$HOME/qemu-6.2.0-rpios/bin:
$PATH"
#export PIPELINEC_REPO=$HOME/PipelineC
export OSS_CAD_SUITE=$HOME/oss-cad-suite/
export PATH="$HOME/.pyenv/bin:$HOME/local/openocd/bin:$PATH"
eval "$(pyenv init --path)"
eval "$(pyenv virtualenv-init -)"
devel@pi5-90:~ $ mkdir spi
```

```
devel@pi5-90:~ $ cd spi
```

```
devel@pi5-90:~/spi $ git clone -b test-dev https://github.com/develone/pico-examples.git
```

Need to modify the files compile-rp2040.sh & compile-rp2040.sh

```
devel@pi5-90:~/spi/pico-examples $ git diff compile-rp2040.sh
diff --git a/compile-rp2040.sh b/compile-rp2040.sh
index 8905c4d..ae7fb2e 100755
--- a/compile-rp2040.sh
+++ b/compile-rp2040.sh
@@ -3,5 +3,5 @@ rm -rf build-rp2040
mkdir build-rp2040
cd build-rp2040
cmake .. -DPICO_BOARD=pico_w -DPICO_PLATFORM=rp2040 -DWIFI_SSID="WIFI_SSID"
-DWIFI_PASSWORD="WIFI_PASSWORD"
-make
+#make
```

```
devel@pi5-90:~/spi/pico-examples $ git diff compile-rp2350.sh
diff --git a/compile-rp2350.sh b/compile-rp2350.sh
index 596732e..af2b30d 100755
--- a/compile-rp2350.sh
```

```

+++ b/compile-rp2350.sh
@@ -3,5 +3,5 @@ rm -rf build-rp2350
mkdir build-rp2350
cd build-rp2350
cmake .. -DPICO_BOARD=pico2_w -DPICO_PLATFORM=rp2350 -
DWIFI_SSID="WIFI_SSID" -DWIFI_PASSWORD="WIFI_PASSWORD"
-make
+#make

```

```

devel@pi5-90:~/spi/pico-examples $ cd build-rp2350/spi/spi_master_slave/spi_master/

```

```

devel@pi5-90:~/spi/pico-examples/build-rp2350/spi/spi_master_slave/spi_master $ make

```

```

devel@pi5-90:~/spi/pico-examples/build-rp2350/spi/spi_master_slave/spi_master $ scp
spi_master.uf2 pi5-70:/media/devel/RP2350

```

```

devel@pi5-90:~/spi/pico-examples/build-rp2350/spi/spi_master_slave/spi_slave $ make

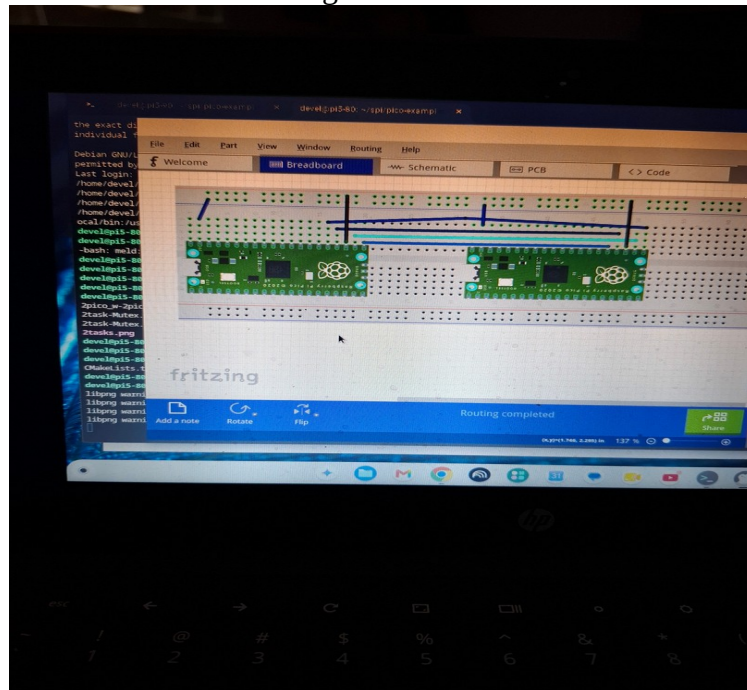
```

```

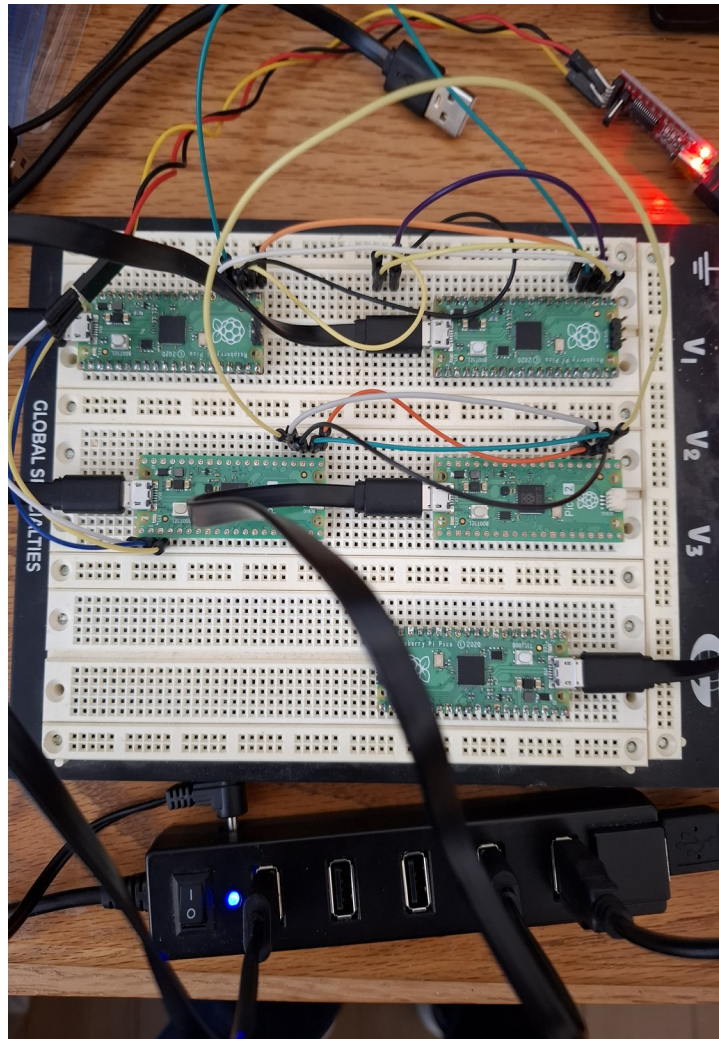
devel@pi5-90:~/spi/pico-examples/build-rp2350/spi/spi_master_slave/spi_slave $ scp spi_slave.uf2
pi5-70:/media/devel/RP2350

```

The master is on the left and slave is on the right.



The pico2 master slave in the middle of the breadboard. The pico master slave on the top.



The shell on left is the spi_master.

// commonly used crc test data and also space for the crc value

```
static uint8_t src[TOTAL_LEN] = { 0x31, 0x32, 0x33, 0x34, 0x35, 0x36, 0x37, 0x38, 0x39, 0x00,
0x00, 0x00, 0x00 };
```

```
crc32 0xd9 0xc6 0xb 0x34
```

```

File Edit Tabs Help
devel@pi5-70: ~
out_buf
0x31 0x32 0x33 0x34 0x35 0x36 0x37 0x38 0x39 0xd9 0xc6 0xb 0x34
in_buf
0x31 0x32 0x33 0x34 0x35 0x36 0x37 0x38 0x39 0xd9 0xc6 0xb 0x34
out_buf
0x31 0x32 0x33 0x34 0x35 0x36 0x37 0x38 0x39 0xd9 0xc6 0xb 0x34
in_buf
0x31 0x32 0x33 0x34 0x35 0x36 0x37 0x38 0x39 0xd9 0xc6 0xb 0x34
out_buf
0x31 0x32 0x33 0x34 0x35 0x36 0x37 0x38 0x39 0xd9 0xc6 0xb 0x34
in_buf
0x31 0x32 0x33 0x34 0x35 0x36 0x37 0x38 0x39 0xd9 0xc6 0xb 0x34

```

```

File Edit Tabs Help
devel@pi5-70: ~
out_buf
0x31 0x32 0x33 0x34 0x35 0x36 0x37 0x38 0x39 0xd9 0xc6 0xb 0x34
in_buf
0x31 0x32 0x33 0x34 0x35 0x36 0x37 0x38 0x39 0xd9 0xc6 0xb 0x34
out_buf
0x31 0x32 0x33 0x34 0x35 0x36 0x37 0x38 0x39 0xd9 0xc6 0xb 0x34
in_buf
0x31 0x32 0x33 0x34 0x35 0x36 0x37 0x38 0x39 0xd9 0xc6 0xb 0x34
out_buf
0x31 0x32 0x33 0x34 0x35 0x36 0x37 0x38 0x39 0xd9 0xc6 0xb 0x34
in_buf
0x31 0x32 0x33 0x34 0x35 0x36 0x37 0x38 0x39 0xd9 0xc6 0xb 0x34

```

The shell on right is the spi_slave.

Next steps adding the preamble.

